Extra notes on F test and t test under SLR (not required)

Under SLR model, suppose that Assumptions (1)-(5) hold. For testing

$$H_0: \beta_1 = 0 \quad \text{v.s.} \quad H_1: \beta_1 \neq 0,$$

at α level, we have the following results:

(i) $t^2 = F$

(ii) $t^2_{n-2;\,\alpha/2} = F_{1,n-2;\,\alpha}$

(iii) The conclusion of t-test and that of F test are always the same (for the same α).

We show (i) & (iii). (ii) needs advanced distribution theory which is not covered here.

Proof of (i):

$$t^2 = \left(\frac{\hat{\beta}_1}{\sqrt{s^2/S_{xx}}}\right)^2 = \frac{(S_{xy}/S_{xx})^2}{s^2/S_{xx}} = \frac{S_{xy}^2/S_{xx}^2}{s^2/S_{xx}}$$

$$= \frac{S_{xy}^2/S_{xx}}{s^2}$$

Recall that $SSR = \dfrac{S_{xy}^2}{S_{xx}}$ and $SSE/(n-2) = s^2$,

so $t^2 = \dfrac{SSR}{SSE/(n-2)} = F$.

$\square$

Proof of (iii)

In t-test we reject H_0 if and only if

$$|t| > t_{n-2;\alpha/2}$$

$$\Longleftrightarrow \quad t^2 > t^2_{n-2;\alpha/2}$$

By (i) $t^2 = F$ and by (ii) $t^2_{n-2;\alpha/2} = F_{1,n-2;\alpha}$,

So $t^2 > t^2_{n-2;\alpha/2}$

$$\Longleftrightarrow \quad F > F_{1,n-2;\alpha}$$

which is the if and if condition for rejecting H_0 using F-test. $\square$